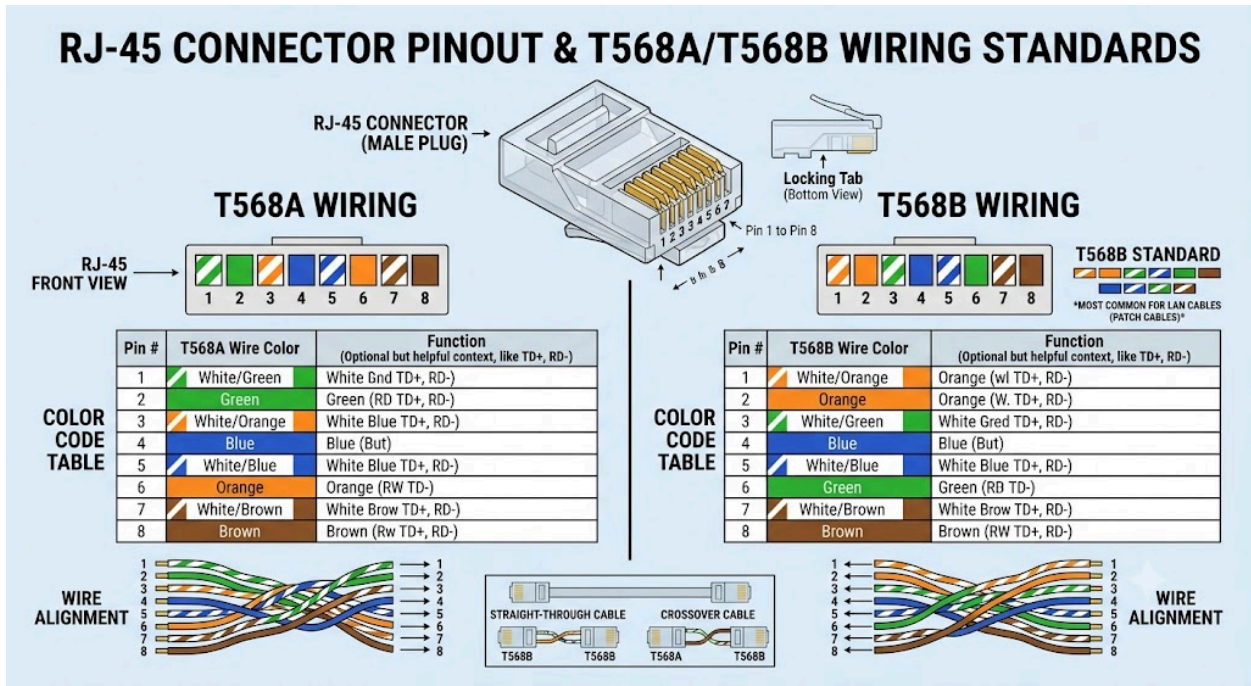


ETHERNET STANDARD & SPECIFICATION

Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

ANS:



Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

ANS:

Specification	10BASE-T	100BASE-TX (Fast Ethernet)	1000BASE-T (Gigabit Ethernet)
Max Speed	10 Mbps	100 Mbps	1 Gbps (1,000 Mbps)
Minimum Cable Type	Cat3 (or higher)	Cat5 (or higher)	Cat5e (or higher)

Twisted Pairs Used	2 pairs (4 wires)	2 pairs (4 wires)	4 pairs (8 wires)
Max Cable Length	100 meters (328 ft)	100 meters (328 ft)	100 meters (328 ft)
Connector Type	RJ-45	RJ-45	RJ-45

Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.
Hint: Think about how transmit and receive pairs are wired in each cable type.

ANS:

To connect two laptops directly via Ethernet without a switch or router in between, you traditionally use a **crossover cable**.

Why a Crossover Cable is Used

- **Pin Alignment:** On a standard laptop NIC (Network Interface Card), Pin 1 and Pin 2 are dedicated **Transmit (TX)** pins, while Pin 3 and Pin 6 are dedicated **Receive (RX)** pins.
- **Avoiding Signal Collision:** A **straight-through cable** connects Pin 1 to Pin 1 and Pin 2 to Pin 2. If you use it directly between two laptops, both computers try to transmit data on the same pins, resulting in no communication.
- **Crossing TX and RX:** A **crossover cable** uses T568A on one end and T568B on the other. This connects Laptop A's Transmit pins (1 & 2) directly to Laptop B's Receive pins (3 & 6), and vice versa, allowing bidirectional communication.

Modern Exception (Auto-MDIX)

Most modern laptops and network adapters feature **Auto-MDIX** (Automatic Medium-Dependent Interface Crossover). This technology automatically detects the connected cable type and electronically swaps the TX and RX signals internally, allowing a standard **straight-through cable** to work directly between two modern laptops as well.

List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

ANS:

Gaming Console / Desktop PC

- **Supported Standard: 1000BASE-T (Gigabit Ethernet)**
- **Why:** Modern gaming systems and motherboards feature Gigabit Network Interface Cards (NICs) to handle high-bandwidth tasks like downloading multi-gigabyte game updates and maintaining low-latency online multiplayer connections.

Network Office Printer / Smart TV

- **Supported Standard: 1000BASE-T or 100BASE-TX (Fast Ethernet)**
- **Why:** Many networked printers and budget Smart TVs utilize 100BASE-TX because 100 Mbps is more than enough bandwidth to buffer 4K video streams (which average ~25–40 Mbps) or print large PDF documents. Higher-end models support 1000BASE-T for faster local file transfers.

College Lab Network Switch / Server

- **Supported Standard: 1000BASE-T (or 10GBASE-T / 1000BASE-LX for uplinks)**
- **Why:** Campus infrastructure switches route high volumes of simultaneous traffic from dozens of workstation nodes, requiring at least Gigabit speeds across all copper access ports to prevent network bottlenecks.